

A Framework for Efficient Search of High-Distance Quantum Stabilizer Codes

Project Synopsis

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Abstract. Quantum stabilizer codes are essential for protecting quantum information, but systematic searches for high-distance codes become computationally prohibitive as code length increases. Existing code constructions exhibit structural regularities, yet no general framework quantifies how such constraints reduce the stabilizer-code search space or incorporates them into code searches. This project asks whether structural properties of known quantum codes can be formulated as mathematical constraints that quantitatively reduce search spaces while remaining useful for discovering high-distance codes. We develop a unified framework based on q -Galois matrices and define the search-space retention factor as the asymptotic fraction of logarithmic search space retained after imposing a constraint. We derive retention factors for weight-spectrum, qudit-participation, overlap, support, Pauli-distribution, local non-commutation, graph-theoretic, and symmetry constraints, including combinations. We introduce finite-size correction factors C and C' and find empirically that they decay approximately as $\log n/n$ and $1/n$, respectively. The framework is evaluated using 58,902 quantum-code parameter sets, including 34,775 codes with usable matrix data across $q = 2, 3, 4, 5, 7, 8$; 1,909 semi-perfect quantum codes are analyzed. We develop a general method to empirically estimate the weight-spectrum and average-weight for unresolved semi-perfect quantum codes, achieving runtime speedups ranging from $10^{8.3608}$ to $10^{193.9298}$ across 4 distinct unresolved codes. Runtime analysis validates that these reductions translate into substantial computational savings. Applying the framework to a $(14, 3)$ qubit search produced a code detecting 91,765 of 91,770 errors of weight at most 4 and achieving 99.999933% and 99.9733% of the theoretically optimal performance of $[[14, 3, 5]]_2$ code at $p = 0.01$. Thus, the framework converts structural properties into explicit candidate-space reductions that can be imposed during code enumeration, providing a quantitative basis for reducing the computational cost of high-distance stabilizer-code searches.

Keywords: Quantum error correction; quantum stabilizer codes; search-space retention factor; quantum code search; finite-size corrections; semi-perfect quantum codes.

1. Rationale and Research Question

A useful quantum computer must correct errors faster than its hardware creates them. Quantum stabilizer codes are the standard tool for this: a code with parameters $[[n, k, d]]_q$ stores k logical qudits in n physical qudits and protects against errors up to a distance d . Larger d means better protection, so finding codes of the highest possible distance for a given n and k is a central open problem.

The obstacle is combinatorial. A stabilizer code on n qudits is specified by an $(n-k) \times n$ matrix over $\text{GF}(q^2)$, so an unconstrained search must consider $q^{2n(n-k)}$ candidates. At $n = 14, k = 3$ this already exceeds 10^{92} configurations, and the cost grows super-exponentially. Exhaustive search is therefore impossible beyond very small codes, and many code parameters remain unresolved decades after they were first tabulated.

Practitioners have long observed that good codes are not random: their stabilizer generators tend to share a fixed weight, overlap in regular patterns, and carry large symmetry groups. These regularities are used informally to guide searches, but they have never been given a common quantitative measure. Without one, there is no principled way to compare two structural assumptions, to predict how much computation an assumption saves, or to decide which assumptions can safely be combined.

Research question. *Can the structural properties of known quantum codes be formulated as mathematical constraints that quantitatively reduce the stabilizer-code search space, while remaining permissive enough to retain high-distance codes?*

Hypothesis. Structural regularities can be encoded as constraints whose asymptotic effect on the search space is captured by a single scalar, and the resulting predicted reductions will be realised as genuine runtime savings in an actual code search.

2. The Framework

2.1 Representation and the retention factor

Each stabilizer generator is written as a q -Galois string $\alpha \in \text{GF}(q^2)^n$, where $\alpha_i = a_i + \omega b_i$ corresponds to the generalised Pauli operator $X^{a_i} Z^{b_i}$. Stacking the $n - k$ generators gives the q -Galois matrix G : rows are generators, columns are physical qudits. Every structural property of interest becomes a statement about G .

The central quantity of this work measures how strongly a constraint prunes the space of admissible G . For a constraint imposed on a family of codes $[[n, k_n, d]]_q$, we define the **search-space retention factor**

$$\mathcal{R}_{[[n, k_n, d]]_q} = \lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O}, \quad S_O = q^{2n(n-k_n)},$$

where S_O and S_R are the unconstrained and constrained search-space sizes. Thus $\mathcal{R}$ is the fraction of the *logarithmic* search space that survives: $\mathcal{R} = 1$ means no asymptotic reduction, and smaller $\mathcal{R}$ means a stronger cut. Because the comparison is made in the exponent, a modest drop in $\mathcal{R}$ corresponds to an enormous reduction in raw candidate count – the theoretical runtime speedup is $S_O^{1-\mathcal{R}}$.

A smaller $\mathcal{R}$ prunes harder but risks excluding the target code, so the design goal is the smallest $\mathcal{R}$ that still retains the desired family.

2.2 Derived retention factors

Table 1 summarises the retention factors derived in the full paper. Three outcomes recur: several constraints give a genuine entropy-limited reduction, several give none at all asymptotically, and a few remain case-dependent. Notably, all the Pauli-distribution constraints yield $\mathcal{R} = 1$, because a random assignment of Paulis to the entries of G already approaches an equal distribution as $n \rightarrow \infty$ – a quantitative reflection of the symmetric roles of X , Y and Z .

2.3 Finite-size corrections

Real searches run at finite n , where the asymptotic $\mathcal{R}$ is not exact. We therefore write, for a single finite code,

$$\mathcal{R} = \mathcal{R}_{[[n, k_n, d]]_q} - C, \quad C = H_{q^2}(\delta_{\max}) - \mathcal{R}, \quad \lim_{n \rightarrow \infty} C = 0,$$

Structural constraint	Retention factor $\mathcal{R}$	Regime parameter
Universal stabilizer (rank + commutation)	$\frac{3+r}{4}$	$r = \lim k_n/n$
Weight spectrum $\text{wt}(g) \in \mathcal{W}$	$H_{q^2}(\delta_{\max})$	$\delta_{\max} = \lim \max(\mathcal{W})/n$
Regular qudit participation	$H_{q^2}(\delta')$	$\delta' = \lim r/(n - k_n)$
Bounded qudit participation	$H_{q^2}(\delta')$	$\delta' = \lim r_{\max}/(n - k_n)$
Symmetric Gram design ($O_{ij} = \lambda$)	$\sqrt{\delta} \log_{q^2}(q^2 - 1)$	$\delta = \lim \lambda/n$
Quasi-symmetric design (λ or μ)	$\sqrt{\delta_{\max}} \log_{q^2}(q^2 - 1)$	$\delta_{\max} = \max(\lim \lambda/n, \lim \mu/n)$
Bounded overlap ($O_{ij} \leq \lambda$)	$\frac{H_q(\sqrt{\delta}) + \sqrt{\delta} \log_q(q+1)}{2}$	valid for $\sqrt{\delta} \leq 1 - q^{-2}$
Symmetric p -wise overlap	$\delta^{1/p} \log_{q^2}(q^2 - 1)$	$\delta = \lim \lambda_p/n$
Bounded p -wise overlap	$\frac{H_q(\delta^{1/p}) + \delta^{1/p} \log_q(q+1)}{2}$	valid for $\delta^{1/p} \leq 1 - q^{-2}$
Non-contained supports	1	no asymptotic reduction
Pauli-distribution constraints	1	no asymptotic reduction
Overlap-graph / automorphism constraints	case-dependent	depends on orbit structure
Local non-commutation	open	needs constrained type enumeration
m simultaneous constraints	$\prod_{i=1}^m \mathcal{R}_i - d \leq \mathcal{R}_{\text{eff}} \leq \min_i \mathcal{R}_i$	$= \prod_i \mathcal{R}_i$ if independent

Table 1: Search-space retention factors derived for each family of structural constraints. H_{q^2} is the q^2 -ary entropy function and d is the clearance constant defined in the full paper. Constraints with $\mathcal{R} = 1$ give no asymptotic saving on their own but may still help at finite n or in combination.

and introduce a second correction $C' = H_{q^2}(\delta_{\text{avg}}) - \mathcal{R}$ in which the maximum relative weight $\delta_{\max}$ is replaced by the *average* relative weight $\delta_{\text{avg}} = w_{\text{avg}}/n$. The behaviour of C and C' is what makes the framework usable in practice, and it is the main empirical subject of the study.

A **semi-perfect quantum code** is defined as a non-degenerate code whose parameters lie on the boundary of the currently known allowable parameter space: no code exists with smaller n , larger d , or larger k when the other two parameters are held fixed. Every perfect code is semi-perfect, but not conversely. These codes are the most structurally regular, and form the central family for the empirical analysis.

3. Materials and Methods

Data. All published code parameters and generator matrices were drawn from `codetables.de` [5] and the Error Correction Zoo [6]: 58,902 parameter sets in total, of which 34,775 carried usable matrix data, spanning field sizes $q = 2, 3, 4, 5, 7, 8$. Within these, all 1,909 known semi-perfect codes were identified and analysed separately.

Correction-factor analysis. For every code, the weight spectrum $\mathcal{W}$ was extracted from its generator matrix, S_R was computed exactly from $(\sum_i \binom{n}{w_i} (q^2 - 1)^{w_i})^{n-k}$, and C (and, for semi-perfect codes, C') was evaluated. Scaling was characterised by fitting the empirical upper envelope $C_{\max}(n) \approx \alpha n^\beta$ per field size, and cross-checked against a Stirling expansion of the exact finite- n expression.

Runtime validation. To test whether a predicted reduction is real, an exhaustive search for the $[[5, 1, 3]]_2$ code was benchmarked under each constraint with early termination disabled, so that the full reduced space is traversed — the worst case in which only one valid configuration exists. Measured speedups were compared against the theoretical $S_O^{1-\mathcal{R}}$.

Application search. The framework was applied to an unresolved target, $(n, k, q) = (14, 3, 2)$.

The predicted weight spectrum was used to decompose generator construction into $|\mathcal{W}|$ sequential stages, each maximising newly detected errors. Stage 1 used local-Clifford and permutation symmetry reduction with an exact Walsh-transform objective and admissible branch-and-bound pruning; later stages added centralizer and quotient-space reduction, Gray-code enumeration, and greedy/beam/LNS search over true error coverage.

4. Results

4.1 The finite-size correction decays as $\log n/n$

Across all 34,775 codes, C decreases steadily with both block length n and dimension k (Figure 1a), confirming that the asymptotic retention factor becomes accurate for longer codes. A Stirling expansion of the exact expression gives

$$C_n = \frac{\log_{q^2}[2\pi n \delta_{\max}(1 - \delta_{\max})]}{2n} - \frac{1}{n} \log_{q^2} \left(\frac{X}{X_{\text{dom}}} \right) + o\left(\frac{1}{n}\right) \implies C = O\left(\frac{\log n}{n}\right),$$

where X is the total combinatorial contribution of all weights and X_{dom} that of the largest weight. Fitting the empirical envelope across field sizes gives $\alpha \ln(q^2) = 0.774 \pm 0.060$ and $\beta = -0.752 \pm 0.023$ ($R_{\log}^2 = 0.98\text{--}0.99$), i.e. $C_{\max}(n) \approx 0.77 \ln(q^2)^{-1} n^{-0.75}$.

The expansion also yields a *parameter-free* bound on the dominant term, $C_{\text{dominant}} \leq \log_{q^2}(\pi n/2)/2n$, which is satisfied by 99.0–99.7% of codes at every field size, with no violation beyond $n = 118$. Two further structural findings emerged: every code with $C < 0$, at every field size, satisfies $\delta_{\max} > \delta^* = 1 - q^{-2}$, so negative corrections occur exclusively beyond the entropy maximum; and below that peak C is predictable from the global parameters (n, k, q) alone with $R^2 \approx 0.948$, falling to $R^2 \approx 0.122$ above it. The practical consequence is that the search-space reduction of a code can be estimated *before* its generators are known.

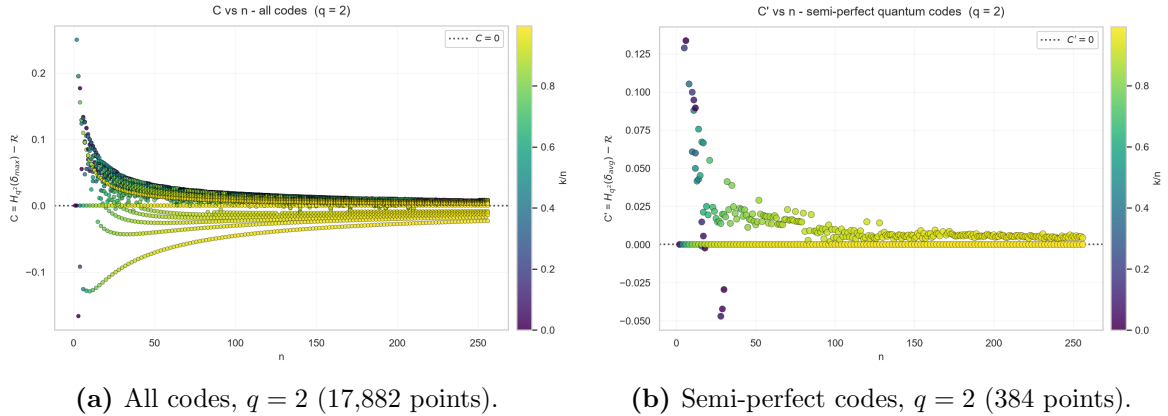


Figure 1: Finite-size corrections at $q = 2$, coloured by code rate k/n . Both vanish as n grows, but the average-weight formulation C' used for semi-perfect codes (b) concentrates far more tightly around zero than the maximum-weight formulation C (a), and decays as $1/n$ rather than $\log n/n$.

4.2 Semi-perfect codes obey a sharper $1/n$ law

Restricting to the 1,909 semi-perfect codes and switching to the average-weight correction C' gives a markedly cleaner result (Figure 1b): the upper envelope is well described by $C'_{\max}(n) \approx b_q/n$, so $C' = O(1/n)$, a strictly faster decay than the $\log n/n$ of the general case. This convergence is specific to semi-perfect codes and is not observed for general quantum codes. The two formulations also converge on one another: $C' = C$ exactly for 1,289 codes

(67.5%), and that fraction rises from 38.5% at $q = 2$ to 92.3% at $q = 8$, with $|C' - C|$ confined to $[-0.0919, 0.0280]$.

This is the result that makes the framework predictive rather than merely descriptive. Because C' is tightly constrained, it can be inverted: given only (n, k, d, q) for a code nobody has yet constructed, one can estimate its average generator weight and hence its full weight spectrum, then use that spectrum as a search constraint. Table 2 applies this to four unresolved semi-perfect codes.

Code	Weight spectrum	$\mathcal{R}$	C'	w_{avg}	Search-space reduction
$[[14, 3, 5]]$	$\{8:3, 10:8\}$	0.9368	0.0530 ± 0.006	9.4582 ± 0.3464	$10^{8.6602}$
$[[17, 3, 5]]$	$\{10:8, 12:6\}$	0.9425	0.0364 ± 0.006	10.9035 ± 0.2845	$10^{13.7075}$
$[[27, 18, 4]]$	$\{18:6, 20:2, 22:1\}$	0.9763	0.0151 ± 0.006	18.4165 ± 0.7502	$10^{8.3608}$
$[[73, 18, 20]]$	$\{40:15, 42:40\}$	0.9260	0.0181 ± 0.006	41.5586 ± 0.5328	$10^{193.9298}$

Table 2: Weight spectra estimated from C' for four unresolved semi-perfect quantum codes, with the resulting reduction in raw search space (equivalently, the runtime speedup available to a search procedure). Spectra are written as {weight : frequency}.

4.3 Predicted reductions are realised as measured speedups

Table 3 benchmarks the $[[5, 1, 3]]_2$ search. The weight-spectrum constraint — the one carried forward into the application — comes closest to its predicted speedup ($\eta = 0.9477$), confirming that the retention factor is a faithful predictor of real computational cost and not merely an accounting identity. Where η falls below 1, the gap is explained by constraint validation overhead, which the theoretical model does not charge for: this is largest for cyclicity, whose runtime (0.227 ms) is so short that fixed setup cost dominates. Since the theoretical speedup grows exponentially in $n(n - k)$ while per-node validation cost stays roughly fixed, this overhead is negligible for the large searches that matter.

Constraint	$\mathcal{R}$	Runtime (s)	Measured speedup	Predicted speedup	Efficiency η
None (unconstrained)	1.0000	13,851.05	1.00 (baseline)	1.00	1.00
Universal stabilizer	0.8469	342.99	40.38	69.69	0.5794
Weight spectrum $\mathcal{W} = \{4\}$	0.8661	361.91	38.27	40.87	0.9477
Cyclicity	0.2500	0.000227	$10^{7.7854}$	2^{30}	0.0568

Table 3: Runtime validation on the $[[5, 1, 3]]_2$ code, with early termination disabled so that the entire reduced space is traversed (worst case). η is the ratio of measured to predicted speedup.

4.4 Application: a new best-known $[[14, 3]]_2$ code

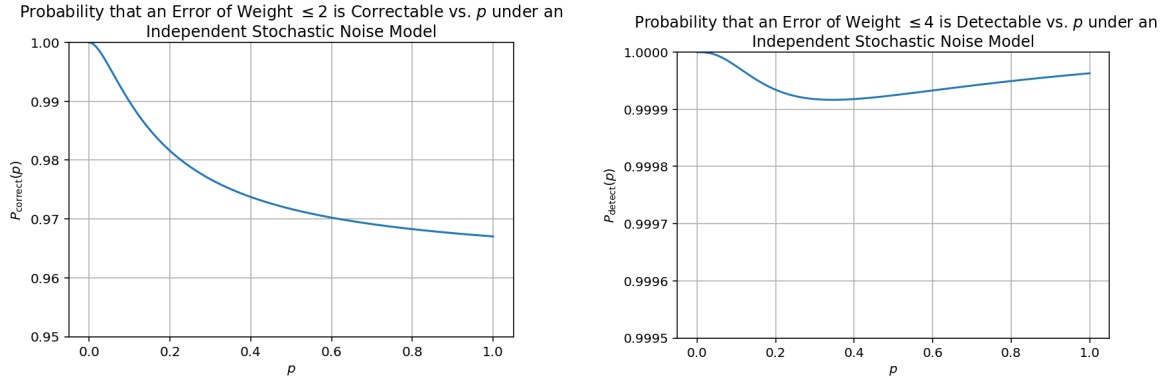
The $[[14, 3, 5]]_2$ code is semi-perfect but unresolved — no explicit construction is known. Taking its estimated average weight 9.4582 ± 0.3464 from Table 2 and comparing neighbouring codes selects the spectrum $\{8:3, 10:8\}$, which reduces the search to a tractable two-stage problem.

Stage 1 (three weight-8 generators) was solved *exhaustively* in 0.26 seconds, yielding 148 optimal column multisets that reduce to 45 inequivalent orbits; each attains the proven Stage-1 optimum, detecting 80,584 of the 91,770 errors of weight ≤ 4 . A complete sweep of Stage 2 over all 45 orbits finished in 19.4 minutes, and seven orbits left only 7 residual undetected

errors. A deeper pass on the orbit with the least-correlated residual profile converged at $t = 1,800.23$ s on a code with just **5** residual errors — the best known $[[14, 3]]_2$ code to date. Its performance:

- detects 91,765 of 91,770 errors of weight ≤ 4 ($\approx 99.995\%$);
- gives distinct syndromes for 808 of 862 errors of weight ≤ 2 (93.74%); the remaining 54 form 27 degenerate pairs, each of which can be assigned a correction strategy, so the code corrects 835 of 862 (96.87%);
- under independent stochastic noise at the hardware-relevant rate $p = 10^{-2}$, achieves $P_{\text{correct}} \approx 0.999734$ and $P_{\text{detect}} \approx 0.99999933$ (Figure 2).

That is, the constructed code reaches 99.9734% and 99.999933% of the correction and detection performance of the theoretically optimal $[[14, 3, 5]]_2$ code at $p = 0.01$. For practical purposes it functions as that optimal code, and it was found in under an hour of total computation on a search space of over 10^{92} candidates.



(a) Correction probability, errors of weight ≤ 2 . (b) Detection probability, errors of weight ≤ 4 .

Figure 2: Performance of the constructed $[[14, 3]]_2$ code as a function of the physical error rate p under an independent stochastic noise model. Both curves stay within a fraction of a percent of unity across the entire range of p , and within 10^{-7} of optimal detection at the hardware-relevant rate $p = 10^{-2}$.

5. Conclusions and Significance

The hypothesis is supported. Structural regularities of quantum stabilizer codes can be written as explicit constraints, each characterised by a single scalar $\mathcal{R}$, and those predicted reductions survive contact with a real search.

1. **A common currency for structural assumptions.** The retention factor lets constraints of completely different character — weight, overlap, symmetry, graph structure — be compared on one scale, and bounded when combined. It also settles which assumptions are worth imposing: the Pauli-distribution constraints, widely used as heuristics, provably give no asymptotic reduction at all.
2. **Scaling laws for finite codes.** $C = O(\log n/n)$ in general and $C' = O(1/n)$ for semi-perfect codes, with a parameter-free bound obeyed by 99% of codes and a sharp regime boundary at the entropy peak $\delta^* = 1 - q^{-2}$.
3. **Prediction before construction.** Because C' is tightly constrained, the weight spectrum of a code that has never been built can be estimated from its parameters alone, converting

an unresolved target into a constrained and tractable search — worth $10^{8.36}$ to $10^{193.93}$ in search-space reduction on the four codes examined.

4. **A concrete new code.** The framework produced the best-known $[[14, 3]]_2$ code, with only 5 residual undetected errors of weight ≤ 4 , matching the detection performance of the optimal $[[14, 3, 5]]_2$ code to within 10^{-7} at realistic error rates.

The framework is independent of the underlying search algorithm, so it can be dropped into backtracking, branch-and-bound, SAT/SMT or ILP searches, with constraints evaluated incrementally as generators are added. This matters because code discovery is currently a bottleneck for fault-tolerant quantum computing: every unresolved entry in the code tables is a gap between what is provably allowed and what anyone can actually build. A method that turns structural intuition into a quantified, reusable search reduction attacks that bottleneck directly.

6. Limitations and Future Work

The retention factor is asymptotic, and although C and C' quantify the finite- n gap, the empirical envelopes are fitted over the block lengths present in the published tables ($n \lesssim 256$). Retention factors for local non-commutation, overlap-graph and automorphism constraints remain case-dependent or open, and the estimates in Table 2 were verified manually rather than by exhaustive construction. Natural next steps are extending the derivations to those open constraint classes and to larger qudit dimensions, integrating the framework directly with SAT/SMT and symmetry-reduction solvers, and applying the weight-spectrum prediction to larger unresolved parameter sets that are currently out of reach.

Author contributions. U.K. conceived the study, developed the methodology, performed the analysis, and wrote the manuscript. **Competing interests.** The author declares none.

Code and data availability. All code and data are publicly available at github.com/utsavkaria-tech. Full derivations, the complete generator matrices, and results for all six field sizes appear in the main paper and its supplementary material.

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